



Basic C++ Program Breakdown (class-object-method)

Objective

Task 1: Create a Car class with:
Data members:
- brand (string),
- model (string),
- year (int)

Methods:
· setDetails() to set car details
· displayDetails() to show car info
· isModern() returns true if year > 2015

Task 2: In main(), create 2 car objects and test all methods.

Class In general

```
class ClassName {  
private:  
    // Data members (attributes/variables)  
    // Only accessible within class  
  
public:  
    // Member functions (methods)  
    // Accessible from outside class  
  
    // Constructor  
    (special method - runs when object is created)  
    internal ClassName() {  
        // Initialization code  
    }  
  
    // Regular methods  
    returnType methodName(parameters) {  
        // Method body  
    }  
};
```

Application

```
#include <iostream>  
#include <string>  
using namespace std;  
  
class Car {  
private:  
    string brand;  
    string model;  
    int year;  
  
public:  
    // Method to set car details  
    void setDetails(string b, string m, int y) {  
        brand = b;  
        model = m;  
        year = y;  
    }  
  
    // Method to display car details  
    void displayDetails() {  
        cout << "Car Details:" << endl;  
        cout << "Brand: " << brand << endl;  
        cout << "Model: " << model << endl;  
        cout << "Year: " << year << endl;  
    }  
  
    // Method to check if car is modern  
    bool isModern() {  
        // Returns true if year is after 2015  
        if (year > 2015) {  
            return true;  
        } else {  
            return false;  
        }  
    }  
};  
  
int main() {  
    // Create two car objects  
    Car car1, car2;  
  
    // Set details for car1  
    car1.setDetails("Toyota", "Corolla", 2018);  
  
    // Set details for car2  
    car2.setDetails("Honda", "Civic", 2010);  
  
    // Display car1 details  
    cout << "=== Car 1 ===" << endl;  
    car1.displayDetails();  
    if (car1.isModern()) {  
        cout << "This is a modern car." << endl;  
    } else {  
        cout << "This is not a modern car." << endl;  
    }  
  
    cout << endl;  
  
    // Display car2 details  
    cout << "=== Car 2 ===" << endl;  
    car2.displayDetails();  
    if (car2.isModern()) {  
        cout << "This is a modern car." << endl;  
    } else {  
        cout << "This is not a modern car." << endl;  
    }  
  
    return 0;  
}
```

Variable Spotlight

```
#include <iostream>  
#include <string>  
using namespace std;  
  
class Car {  
private:  
    string brand;  
    string model;  
    int year;  
  
public:  
    // Method to set car details  
    void setDetails(string b, string m, int y) {  
        brand = b;  
        model = m;  
        year = y;  
    }  
  
    // Method to display car details  
    void displayDetails() {  
        cout << "Car Details:" << endl;  
        cout << "Brand: " << brand << endl;  
        cout << "Model: " << model << endl;  
        cout << "Year: " << year << endl;  
    }  
  
    // Method to check if car is modern  
    bool isModern() {  
        // Returns true if year is after 2015  
        if (year > 2015) {  
            return true;  
        } else {  
            return false;  
        }  
    }  
};  
  
int main() {  
    // Create two car objects  
    Car car1, car2;  
  
    // Set details for car1  
    car1.setDetails("Toyota", "Corolla", 2018);  
  
    // Set details for car2  
    car2.setDetails("Honda", "Civic", 2010);  
  
    // Display car1 details  
    cout << "=== Car 1 ===" << endl;  
    car1.displayDetails();  
    if (car1.isModern()) {  
        cout << "This is a modern car." << endl;  
    } else {  
        cout << "This is not a modern car." << endl;  
    }  
  
    cout << endl;  
  
    // Display car2 details  
    cout << "=== Car 2 ===" << endl;  
    car2.displayDetails();  
    if (car2.isModern()) {  
        cout << "This is a modern car." << endl;  
    } else {  
        cout << "This is not a modern car." << endl;  
    }  
  
    return 0;  
}
```

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Author: Mahtamun Hoque Fahim

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